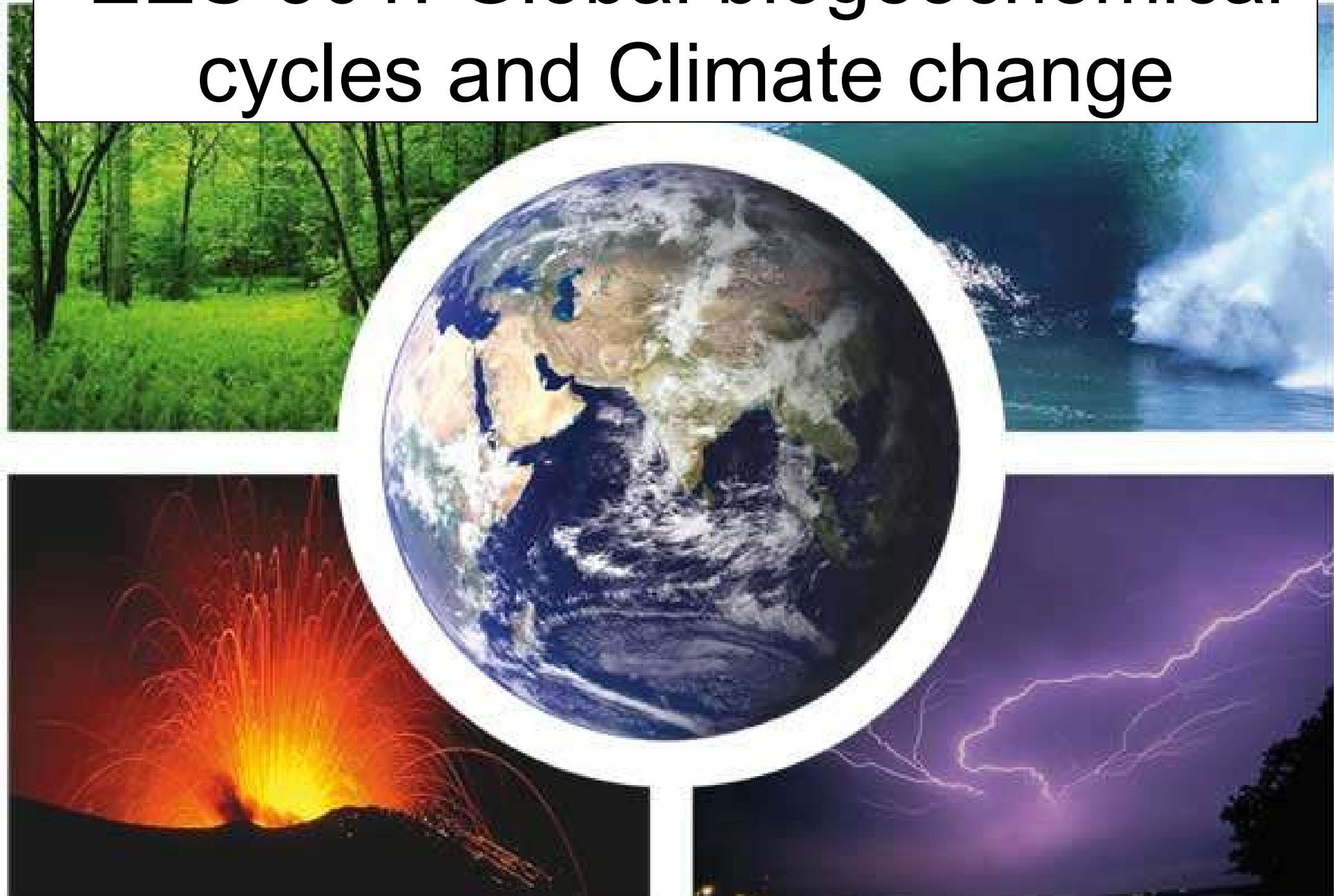


EES 301: Global biogeochemical cycles and Climate change



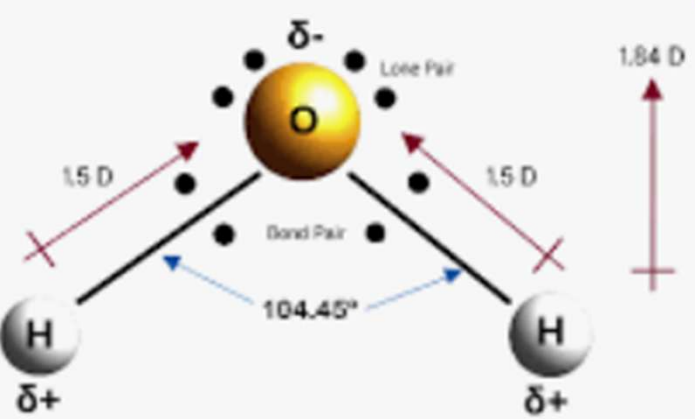
The Hydrological cycle

Earth and Environmental Sciences have a sub discipline, hydrology which focuses on the global hydrologic cycle and the processes involved in the land phase of that cycle. Hydrology describes and predicts:

- 1. The spatial and temporal variations of water substance in the terrestrial, oceanic, and atmospheric compartments of the global water system.**
- 2. The movement of water on and under the earth's surface, the physical, chemical, and biological processes accompanying that movement.**

Water a special substance

The permanent dipole is so strong that it results in the formation of hydrogen bonds between the negative dipole O atom of one molecule and an H atom with positive dipole of a neighboring molecule. The strength of the bond is 20 KJ/mol



Water is a bend molecule with a very strong permanent dipole

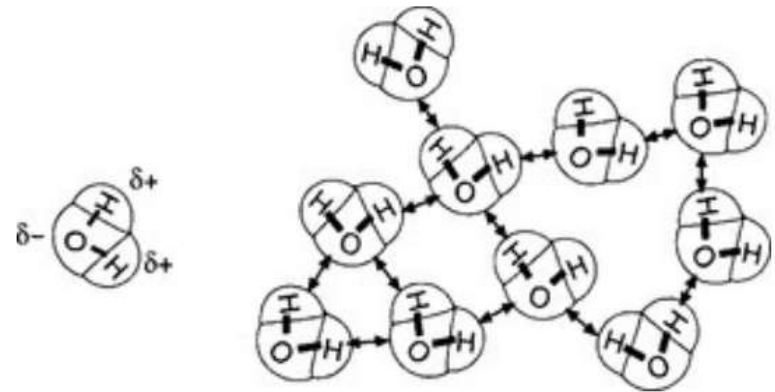


FIG. 6-1 Hydrogen bonds in liquid water.

Water a special substance

These hydrogen bonds give rise to unique chemical and physical properties such as:

- High boiling and melting point
- high surface tension, which among other things makes rain without aerosol impossible
- high heat capacity ($75 \text{ J mol}^{-1} \text{ K}^{-1}$) which permits water to absorb a lot of energy from the sun during the day and release it at night without much change in the temperature of the water
- A high latent heat of evaporation and freezing
- A density maximum at 4°C



Properties of water

Table 6-1

Anomalous properties of water

Property	Water	Comparison species	Heat capacity of liquid	4218 J/(K kg)	CH ₄ :2170 J/K kg
Boiling point	373 K	CH ₄ : 112 K (comparable size molecule)	Latent heat of vaporization (0°C)	2.5 × 10 ⁶ J/kg	
		NH ₃ : 240 K	Latent heat of freezing	3.3 × 10 ⁵ J/kg	
		H ₂ S: 211 K (dihydride)	Ratio of density frozen/density liquid (0°C)	0.92	
		H ₂ Se: 231 K (dihydride)	Surface tension	73 dyn/cm	CCl ₄ : 27 (nonpolar liquids)C ₆ H ₆ : 29
		Ar: 87 K			
Melting point	273 K	CH ₄ : 89 K			
		NH ₃ : 195 K			
		H ₂ S: 190 K			
		H ₂ Se: 209 K			
		Ar: 84 K			

Clausius-Clapeyron equation

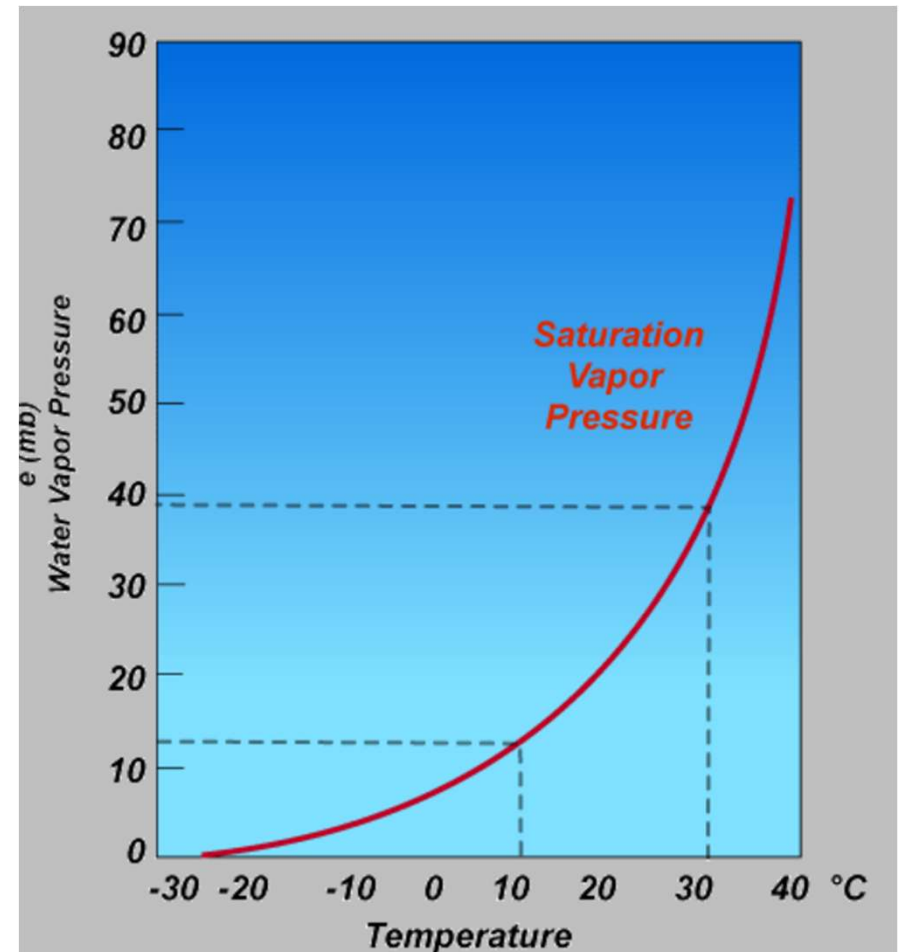
The Clausius-Clapeyron equation describes the dependence of the saturation vapor pressure of water in temperature (T), pressure (P) and the latent heat of evaporation (L_e)

$$\frac{d(\ln p)}{dT} = \frac{L_e}{RT^2}$$

In integral form and between two locations with P_1, T_1 and P_2, T_2 it is written as:

$$\ln \frac{p_2}{p_1} = -\frac{L_e}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

In the context of daily weather you may be more familiar with the concept of **relative humidity (RH)**. The RH is the ratio between the **actual water vapor pressure** and the **saturation vapor pressure of water** at that Temperature expressed in %



Water stability in the atmosphere

Due to the high bond energy of the O-H bond in water (456 kJ/mol) water can not photolyse in the atmosphere.

How to calculate what energy of light is needed to break a certain bond:

Step 1: Convert the Energy from J per mol to J per molecule

$$456000 \text{ J/mol} / 6.022 \times 10^{23} = 7.57 \times 10^{-19} \text{ J per molecule}$$

Step two use the relationship $E = h\nu = hc/\lambda$

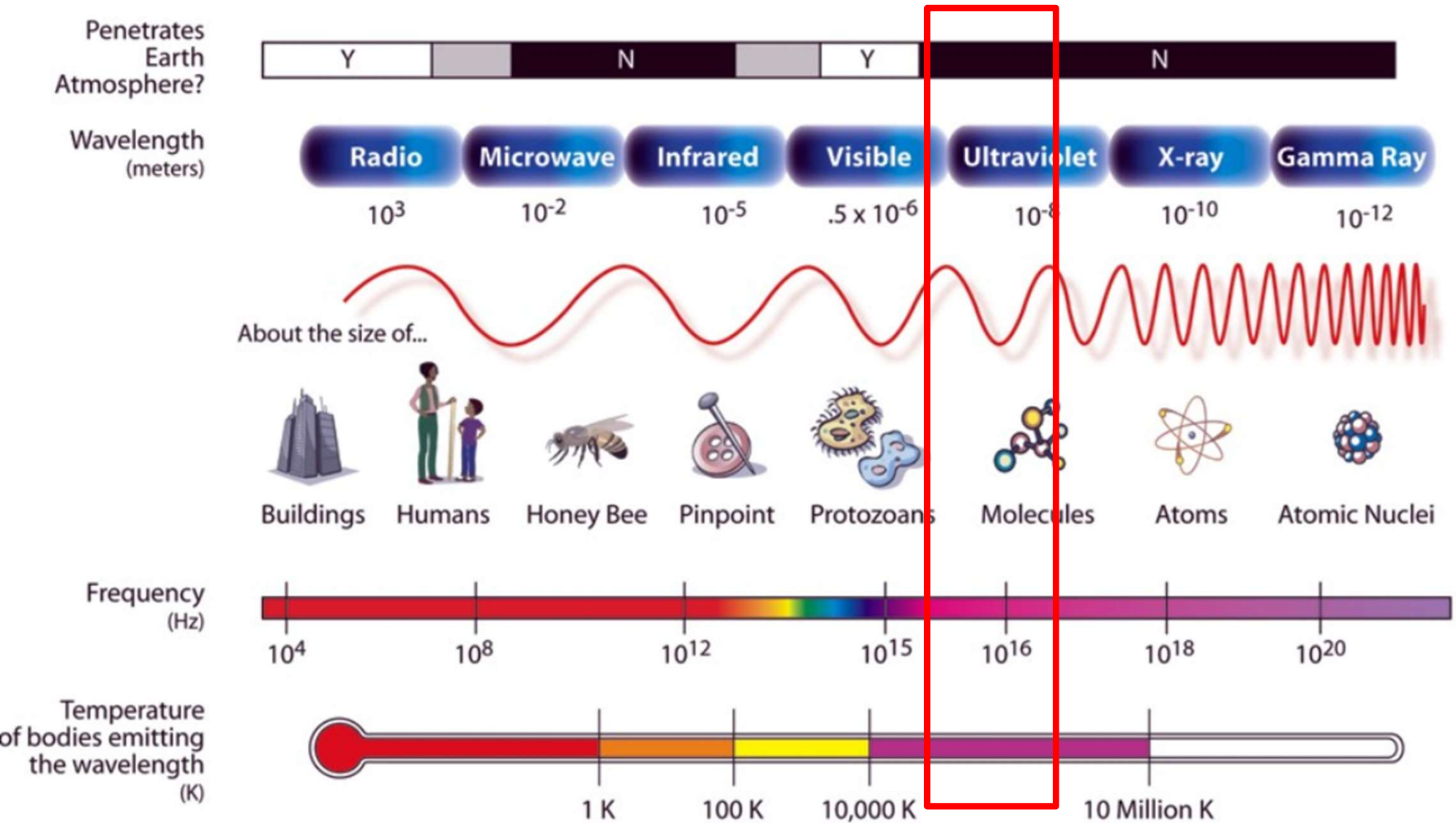
With the Planck constant $h = 6.62607015 \times 10^{-34} \text{ [J}\cdot\text{Hz}^{-1}]$ or $\text{[J}\cdot\text{s]}$ and the speed of light in vacuum $c = 299792458 \text{ m s}^{-1}$

For conversion of water into an OH radical:

$$\lambda = 6.62607015 \times 10^{-34} \text{ [J}\cdot\text{s}] \times 299792458 \text{ m s}^{-1} / 7.57 \times 10^{-19} \text{ [J}^{-1}] = 265 \text{ nm}$$

For conversion into Ion it becomes $<130 \text{ nm}$

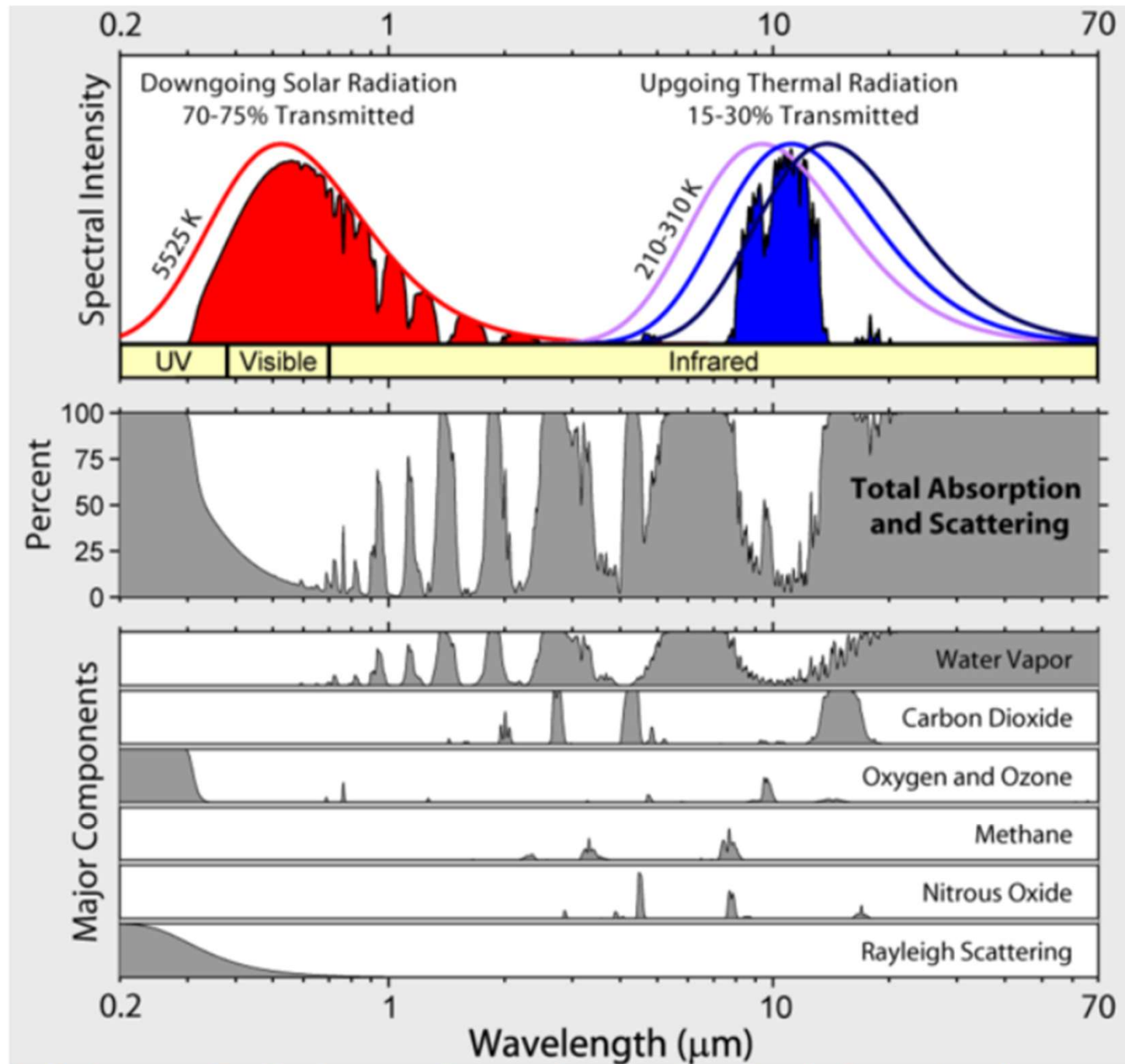
Water stability in the atmosphere



Tropospheric water could photolyze in the early atmosphere before we had an ozone layer, but now it no longer can.

Water as GHG gas in the atmosphere

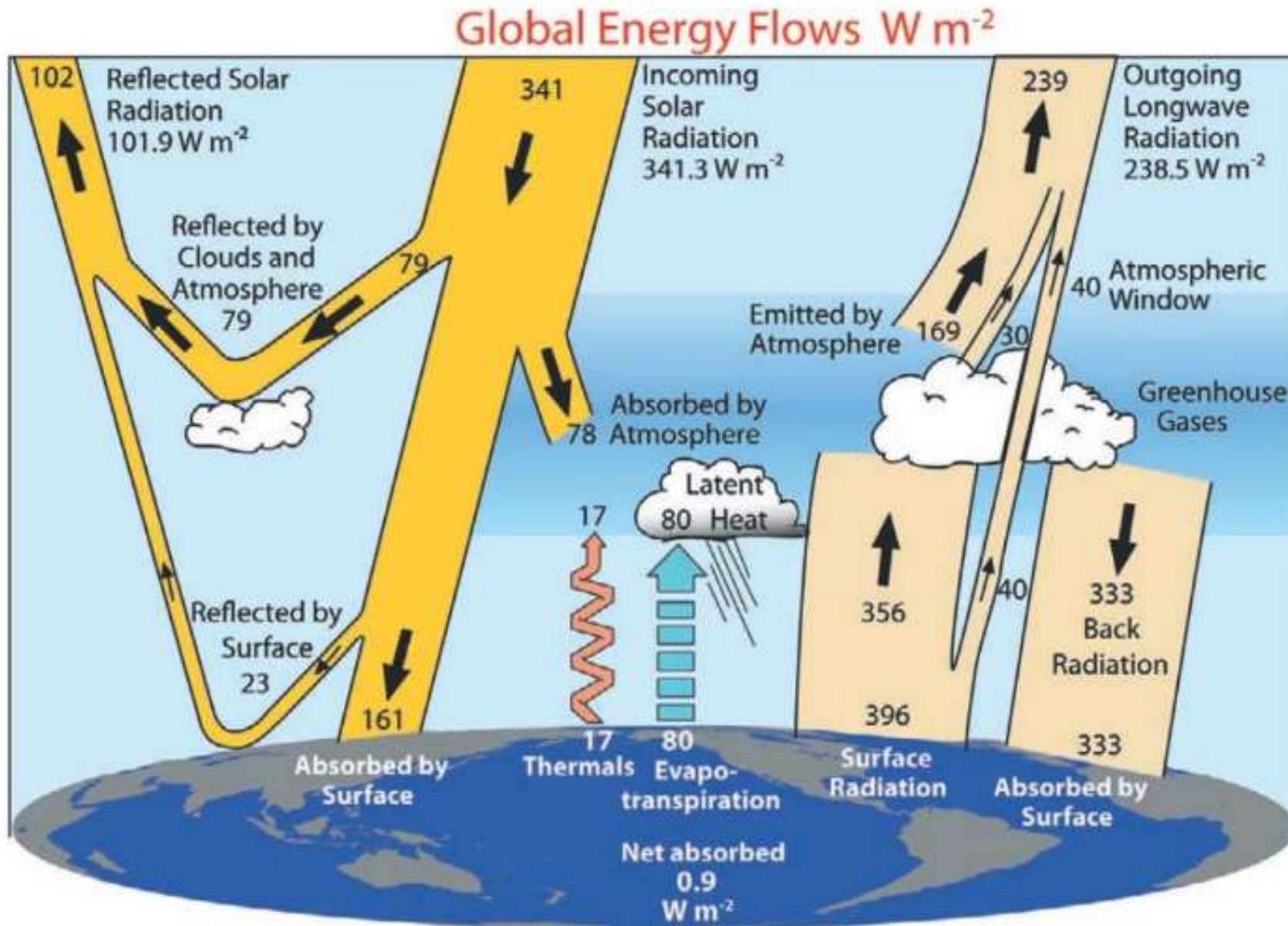
Water absorbs most of the wavelengths emitted by the black body called Earth baring a small window. Making it the most effective Greenhouse gas in the atmosphere



Radiation transmitted by the atmosphere as a function of wavelength. Downgoing solar radiation is on the left, and upgoing thermal radiation is on the right, with the percentage of radiation absorbed and scattered by the Earth's atmosphere below. Finally, the percentage of radiation absorbed and scattered is then divided by the major greenhouse gases in the panels at the bottom ([CC BY-SA 3.0](#)).



Water as GHG gas in the atmosphere



Thanks to the natural Greenhouse effect of H_2O (g) in our atmosphere, the average temperature on Earth is a balmy 15°C rather than a freezing -18°C

The global hydrological cycle

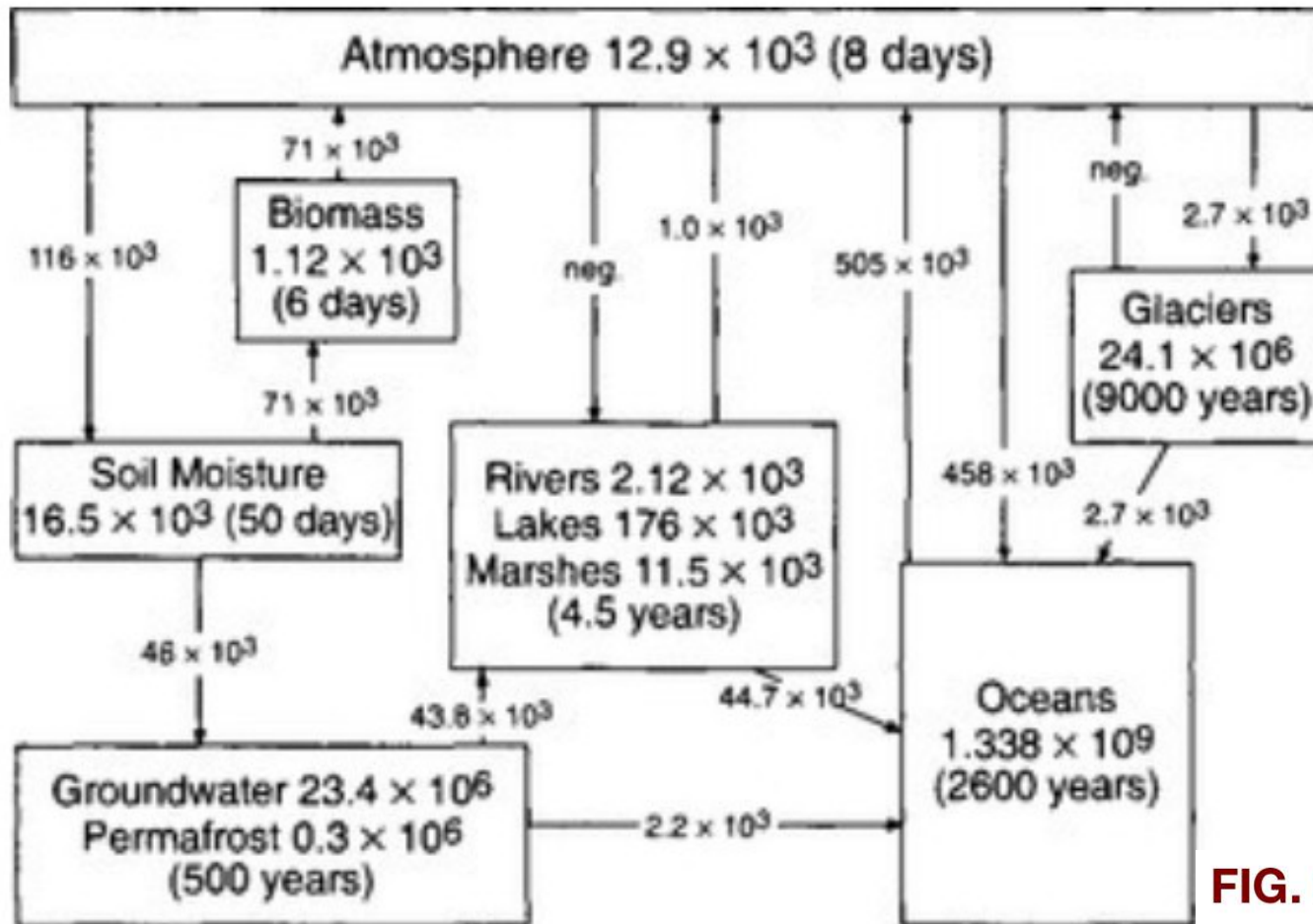
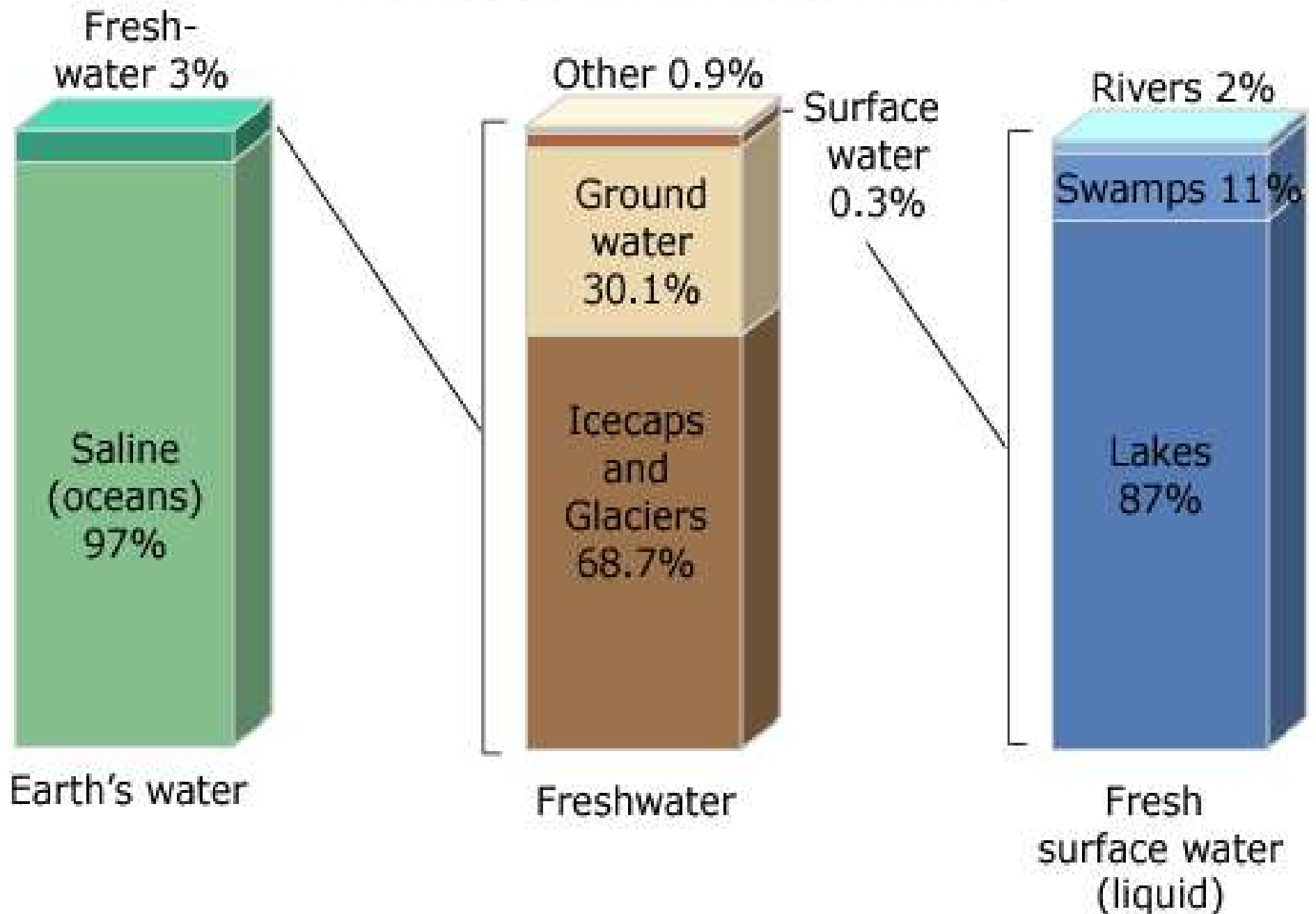


FIG. 6-3 Global water balance. Storages in km^3 , fluxes in km^3/yr . Turnover times calculated as storage divided by total annual inflow.

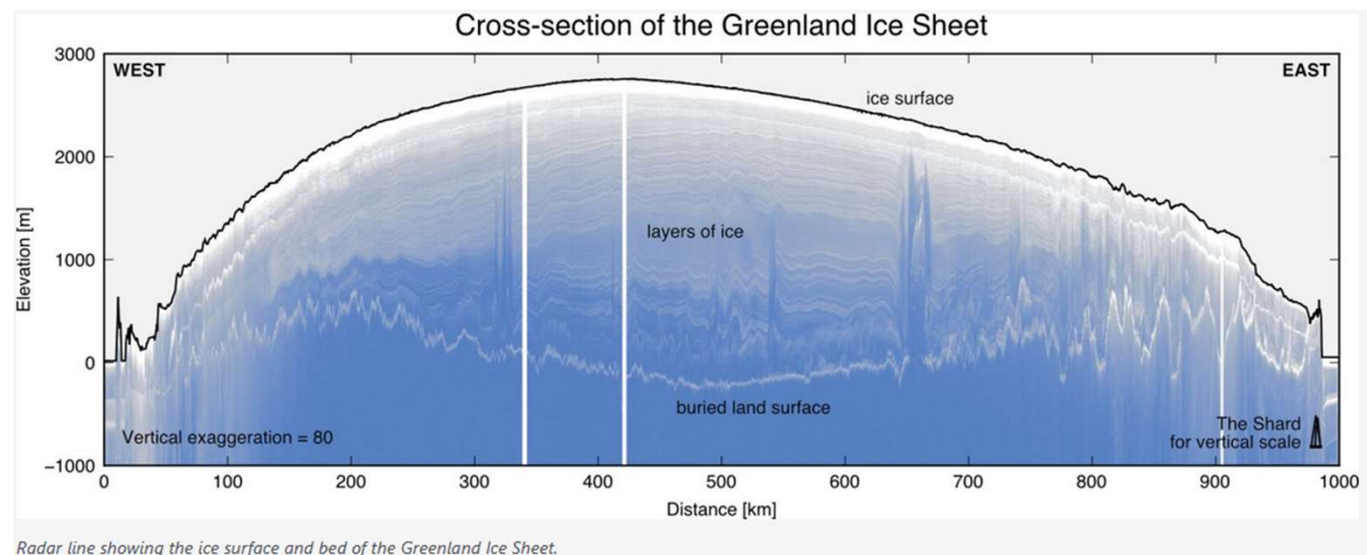
(Data from [Shiklomanov and Sokolov, 1983](#) .)

Distribution of Earth's Water



Constraining the total water content of reservoirs is hard

- Depending on the reservoir we are talking about it is a 3D problem (lateral extend and depth) or a 4D problem (with a time dimensions) e.g. soil moisture.

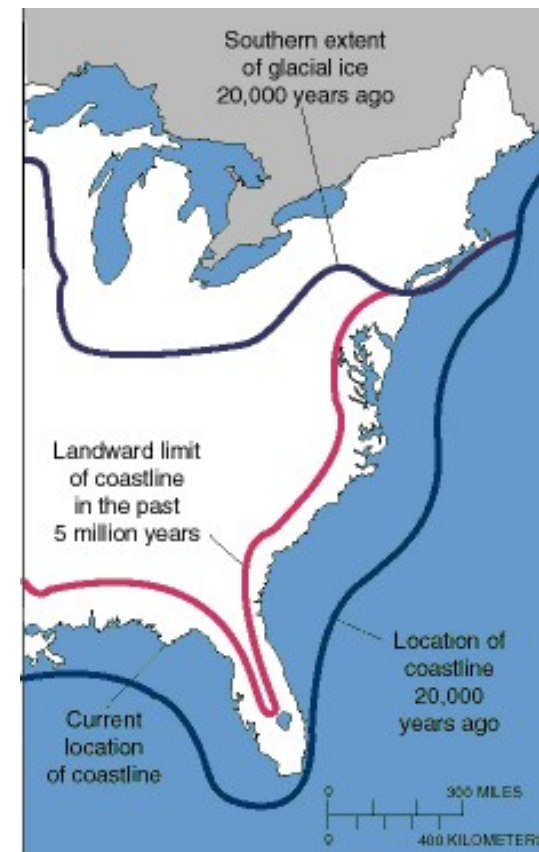
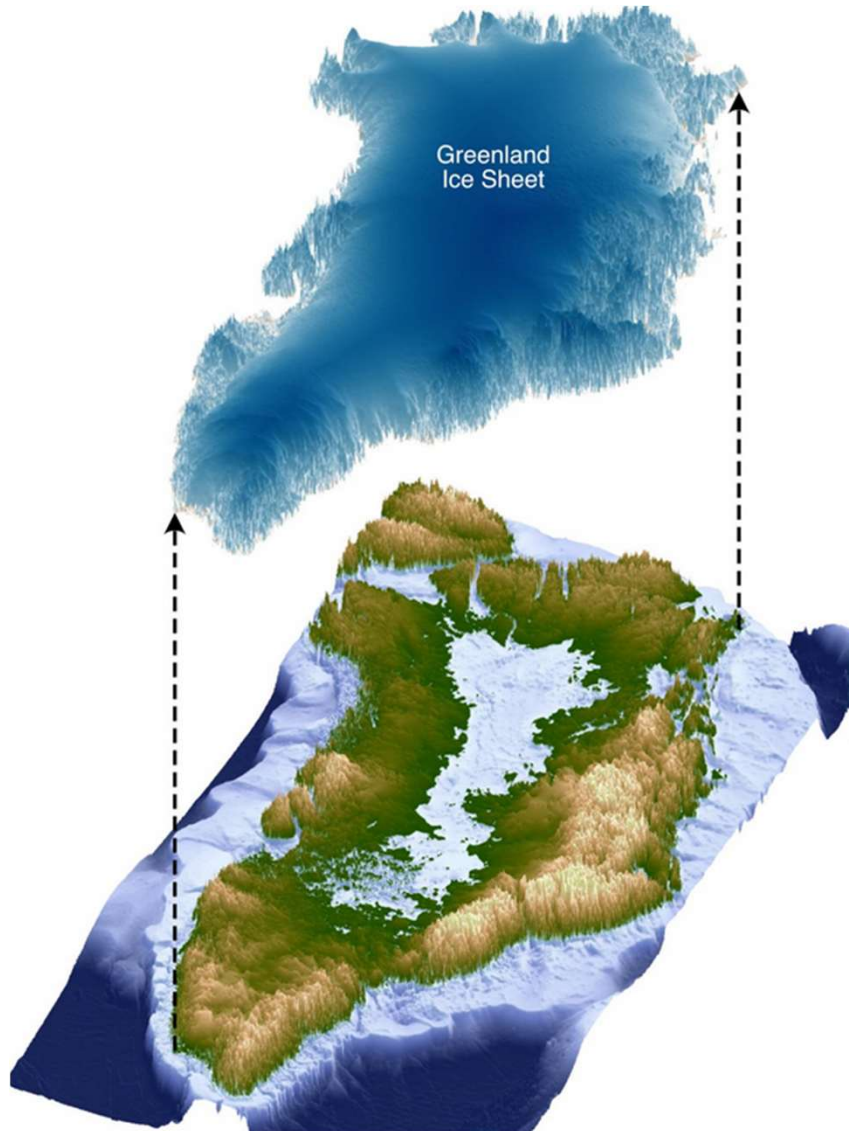


There is an unknown landscape underneath the ice which scientists have recently started to study with ice penetrating radars.

Only after the end of world war 2 did the technology to study the bathymetry of the ocean become available to scientists

Constraining the total water content of reservoirs is hard

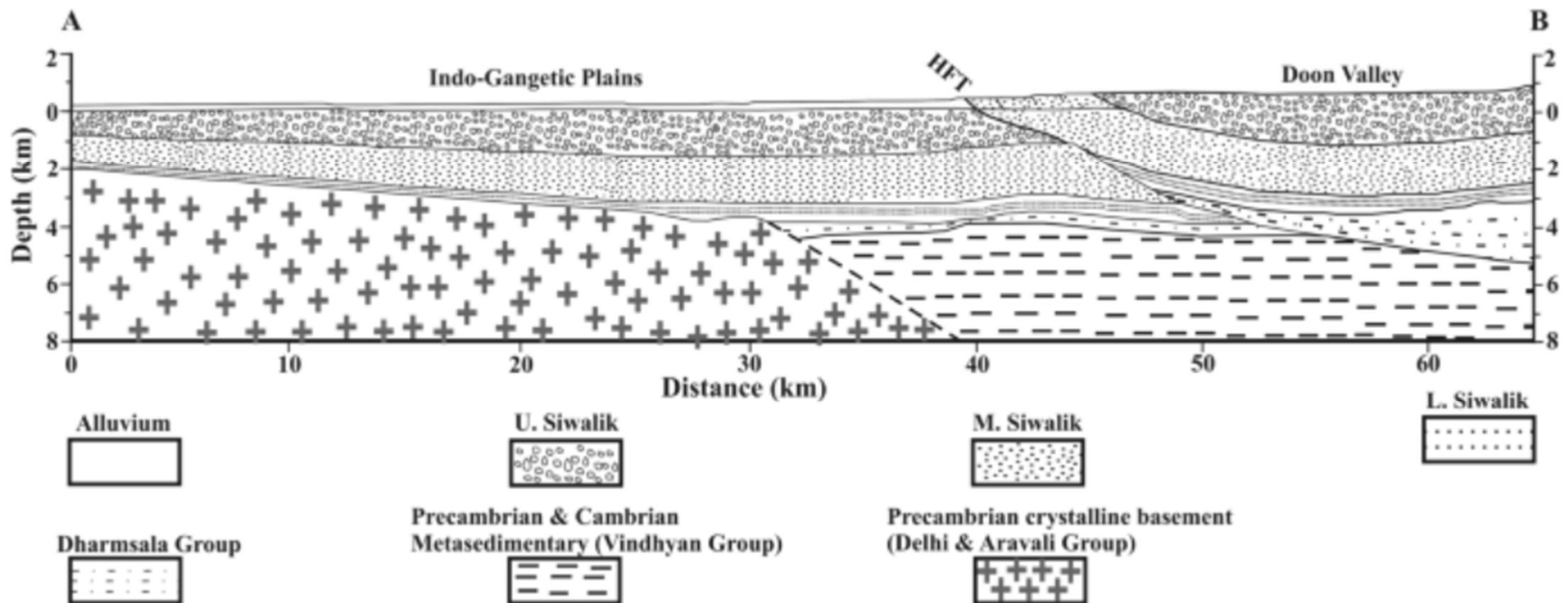
- To make matters more complicated large ice sheets push the crust into the mantle and then it rebounds once the ice melts.
- This complicates the assessment of sea level changes due to melting and mitigates the sea level rise for the regions that had the ice cover.



Constraining the total water content of reservoirs is hard

- Sedimentary basins in subduction zones can contain very thick sedimentary columns in the IGP the column is up to 4 km thick.
- Till what depth these contain groundwater and how much is difficult to assess

Figure 2



Different estimates of water storage in different reservoirs

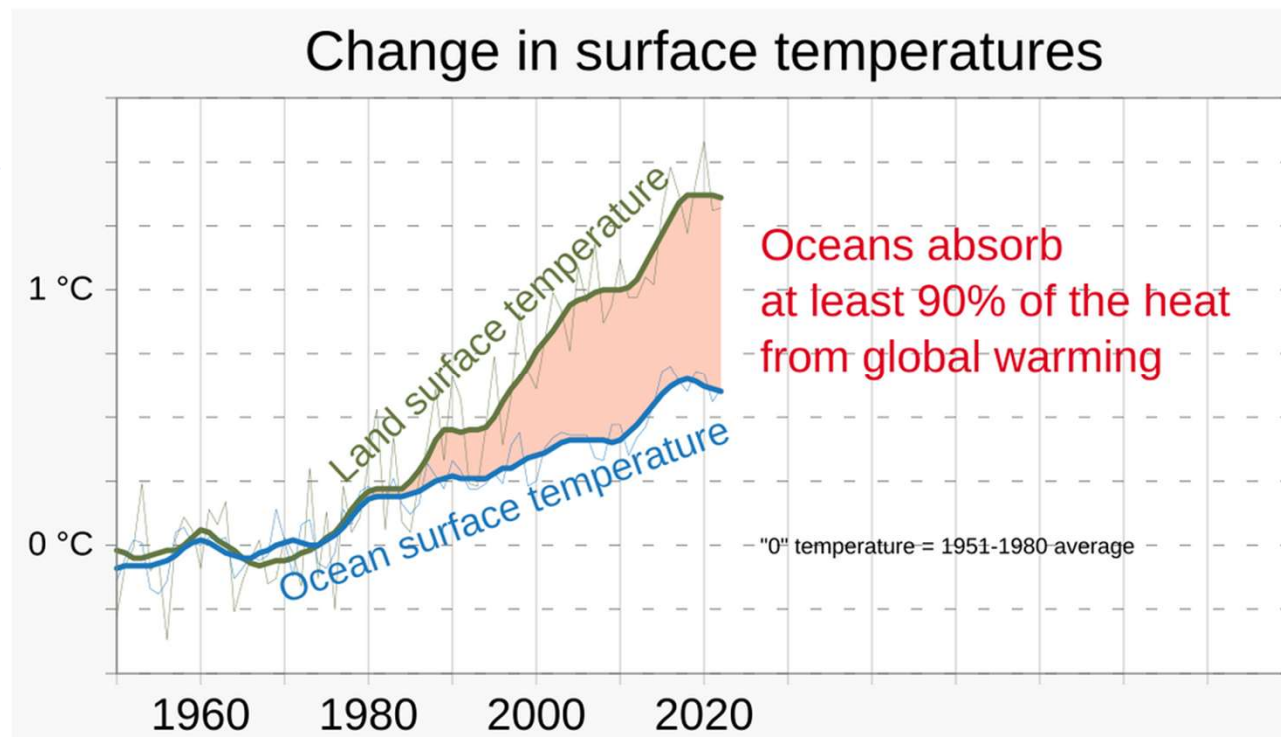
Reservoir storage

UNESCO is far more optimistic on groundwater storage than other authors

Reservoir	UNESCO (1978) (millions of km ³)	Nace (1969) (millions of km ³)	Baumgartner and Reichel (1975) (millions of km ³)
Oceans	1338	1350	1348
Cryosphere	24.06	26.0	27.82
Groundwater	23.40	7.0	8.062
Fresh groundwater	10.53	—	7.956
Soil moisture	0.0165	0.150	0.0612
Permafrost	0.3000	—	—
Lakes	0.1764	0.230	0.224
Freshwater lakes	0.0910	0.125	0.126
Rivers	0.00212	0.0017	0.0011
Marshes	0.01147	—	—
Atmosphere	0.0129	0.013	0.013
Biosphere	0.00112	negligible	0.0011
Total	1386	1384	1384
Freshwater total	35.03	36.0	36.00

The oceans as reservoirs

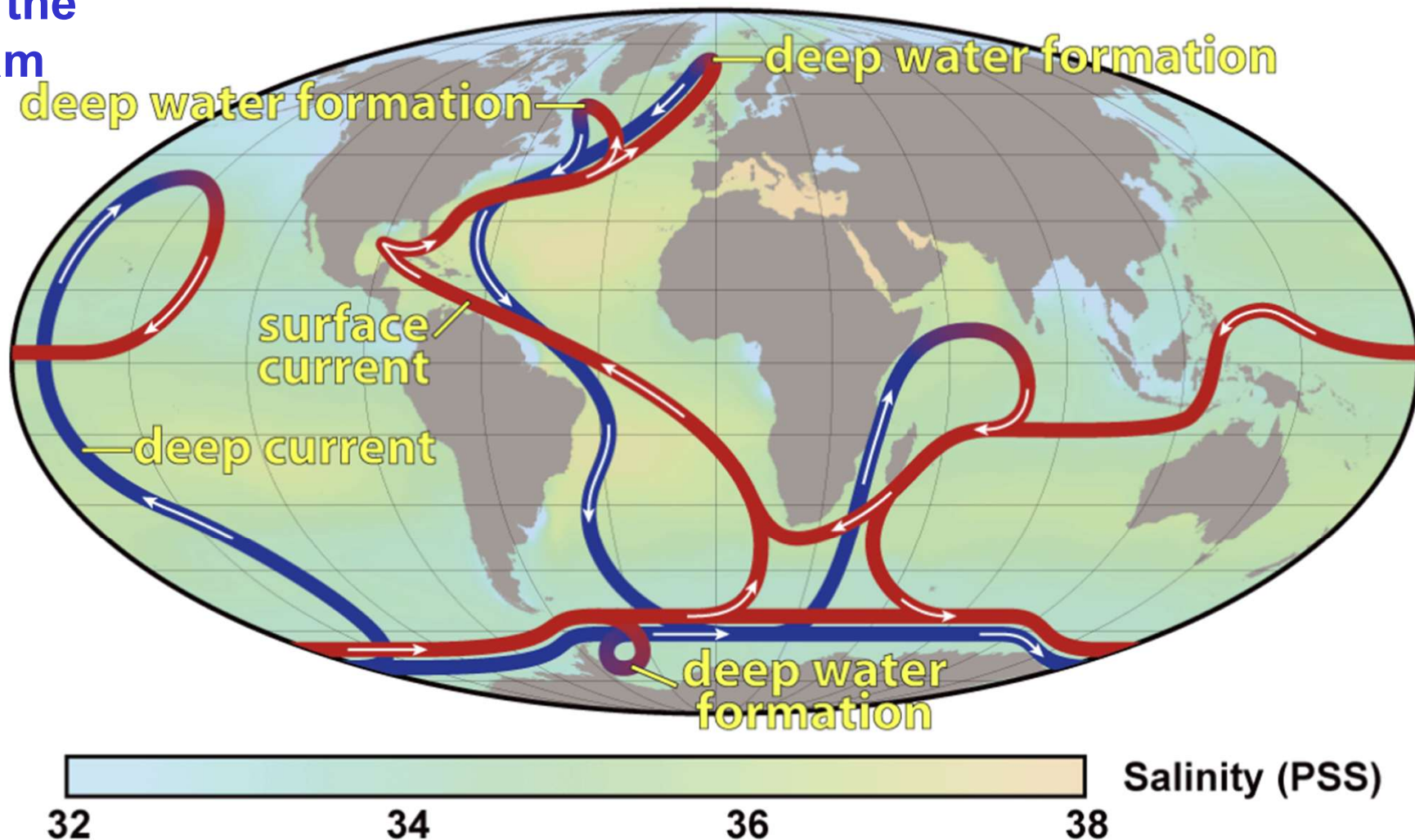
- The largest reservoir containing 97% of the total liquid water on earth total 1338 million km³ of water.
- Oceans loose more water through evaporation than they gain through precipitation resulting in a net transport of water from the ocean to the land and the cyrosphere.
- The oceans have absorbed 90% of the total extra heat retained due to anthropogenic climate change and roughly 50% of the CO₂ emitted so far.
- Thanks to the high heat capacity of water oceans have played a large role in stabilizing Earth's climate over time



The oceans as heat transport engines

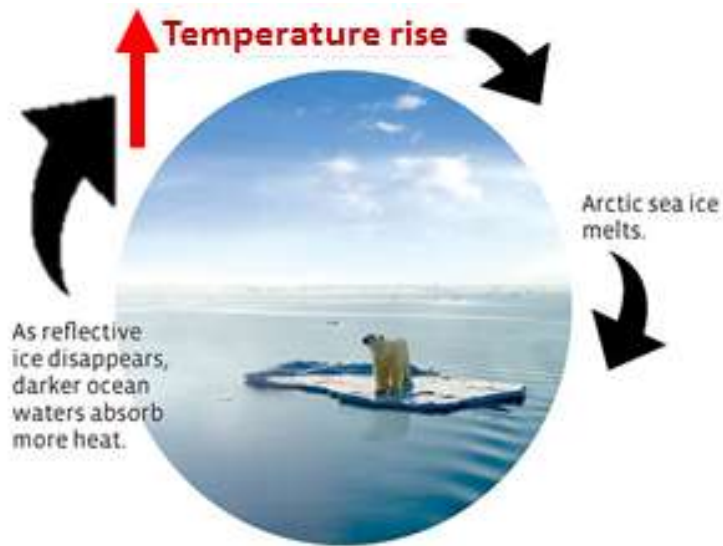
- Oceans redistribute heat from the equator to the pole and into the deep ocean and modulate global temperatures
- The mild climate in Europe is thanks to the Gulf stream

Thermohaline Circulation



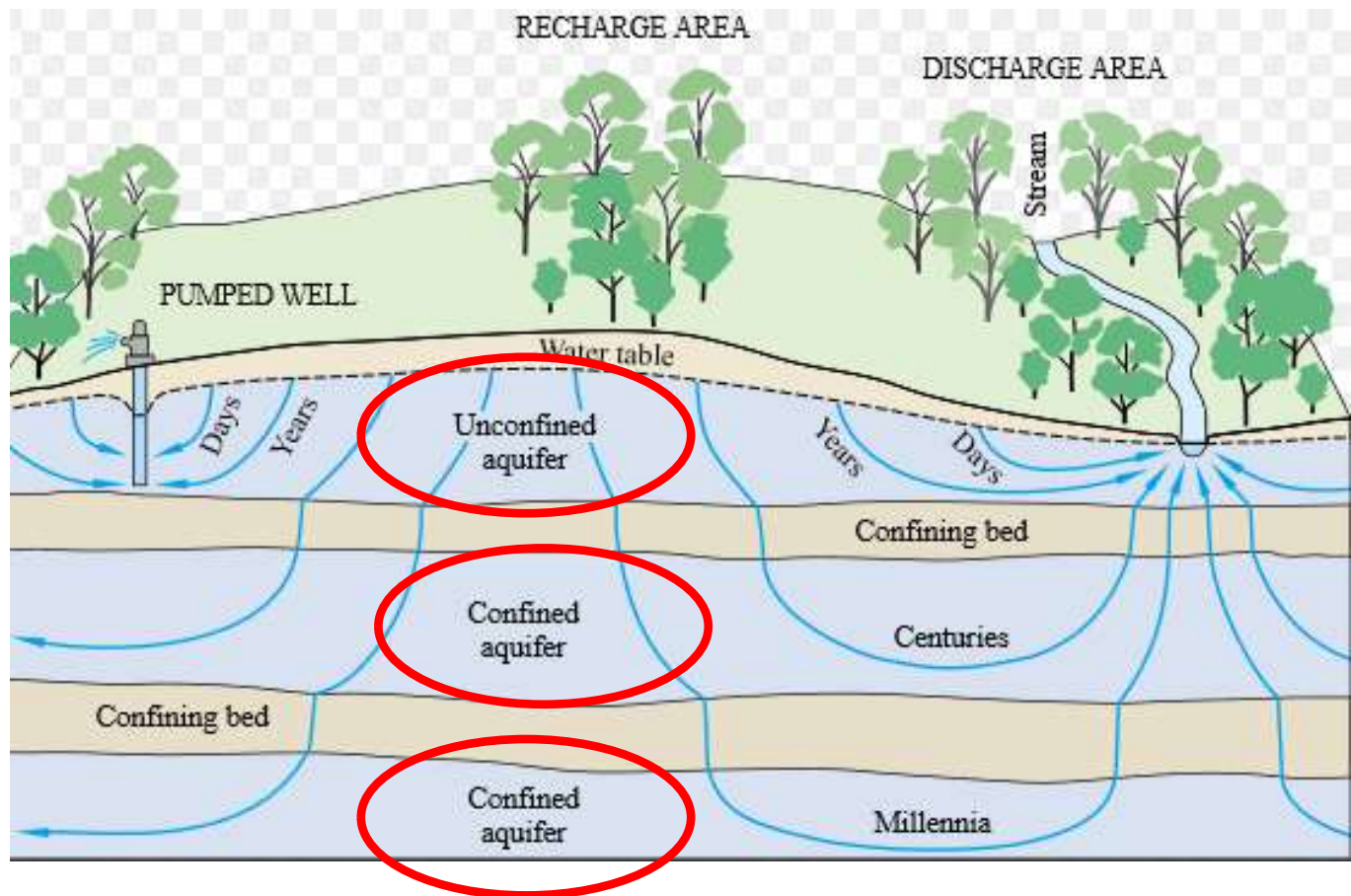
The Cryosphere as the largest freshwater reservoir

- Approximately 68% of the total freshwater roughly 24 to 27 million km³ of water is frozen into ice sheets.
- Most of this water is in remote locations and doesn't help to address freshwater shortages
- Sea ice contributes significantly to the Earth's albedo and its melting amplifies climate change



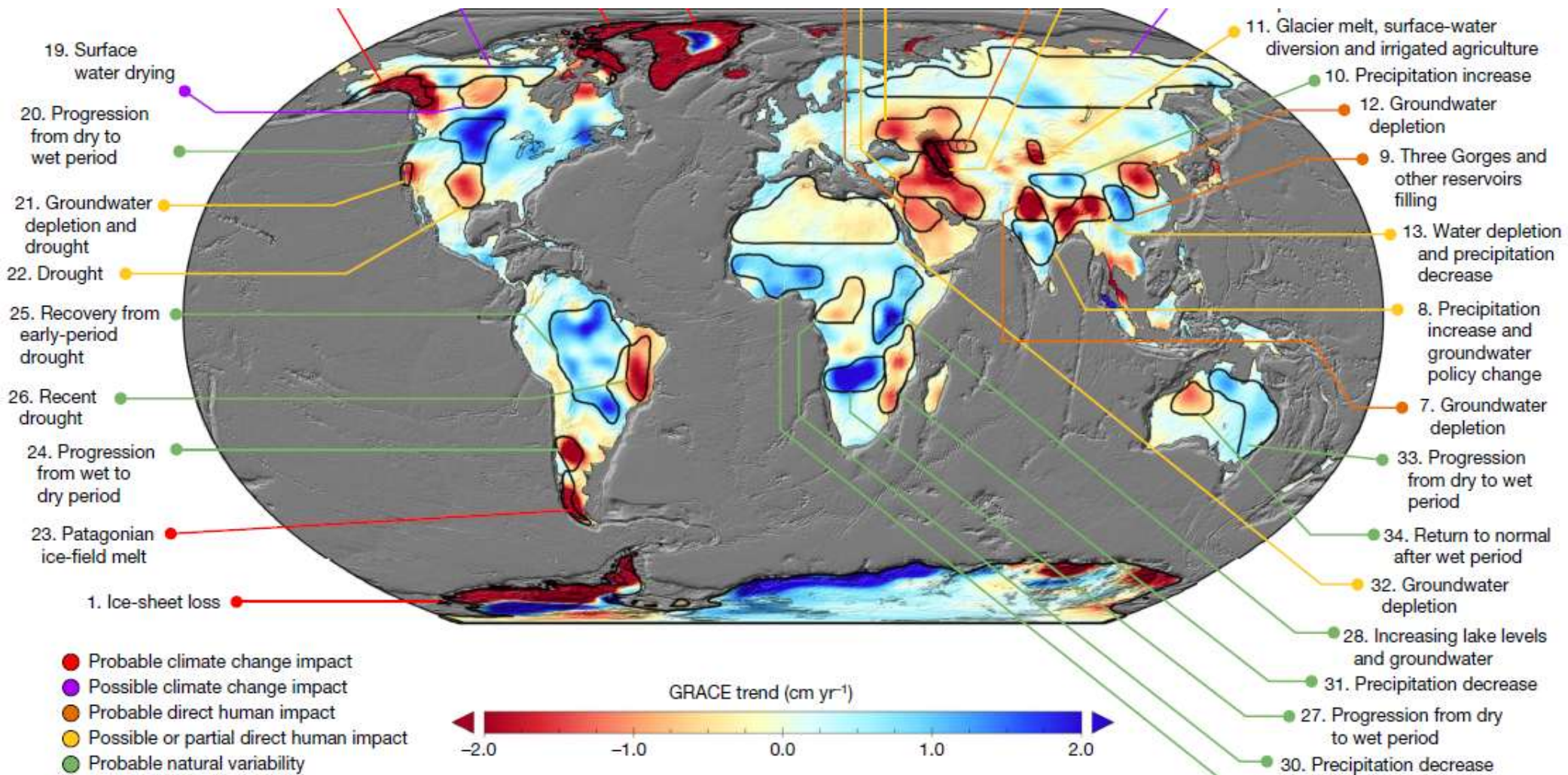
Groundwater – the most used freshwater reservoir

- Approximately 30% of the total freshwater roughly 7 to 23 million km³ of water is present as groundwater in aquifers
- Aquifers are porous rocks or sediments in which the pore spaces are completely filled with water
- The unconfined aquifer is fed by rain via the hydrological cycle and connected to streams. The water here is a renewable resource
- The confined aquifers are separated from layers above by an impermeable rock or sediment. Rain does not feed the confined aquifer. The water in here is century to millennia old. This water is a non-renewable resource



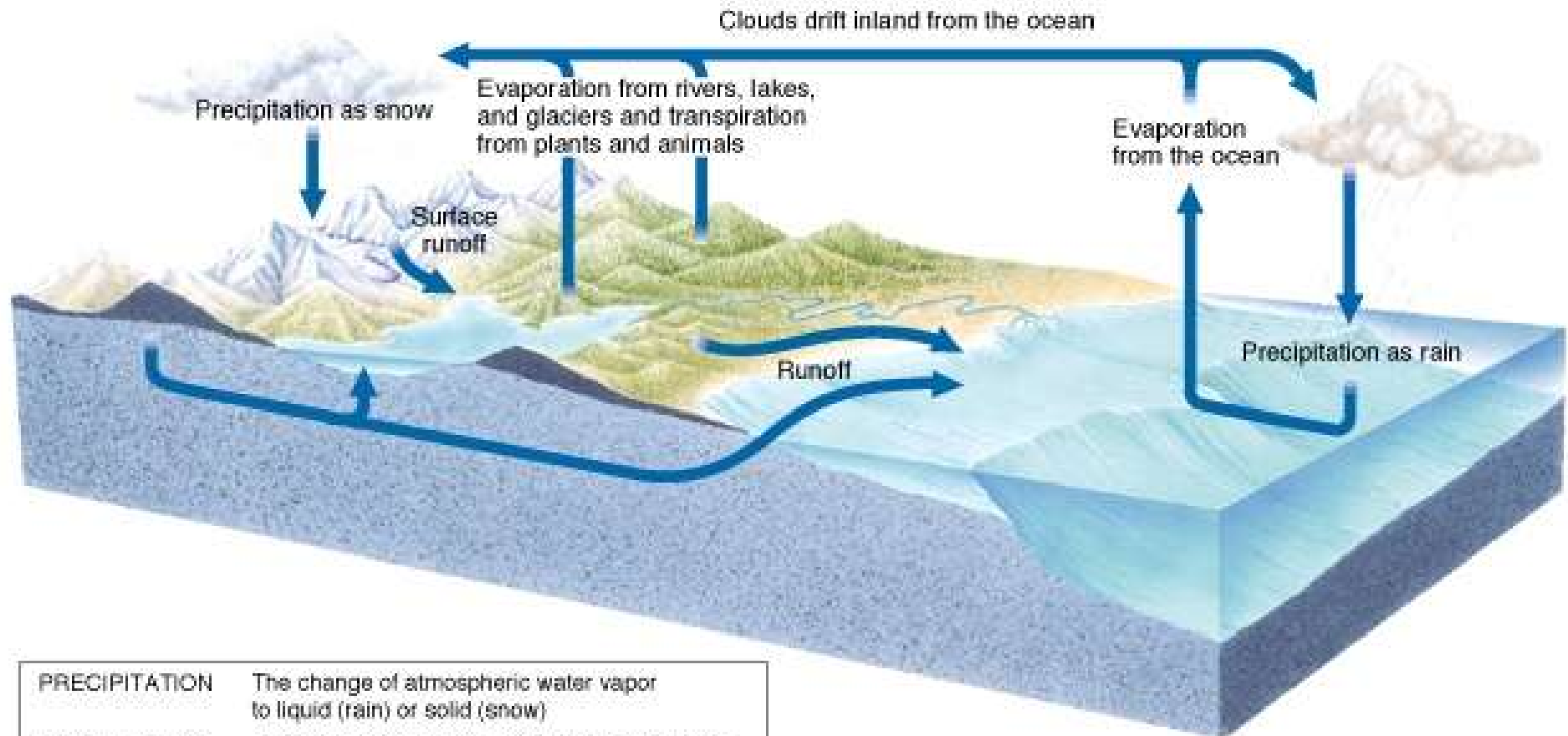
Groundwater depletion

- While it is very hard to determine how much groundwater there is in total, it is much easier to determine by how much we are overusing it. We have a satellite called GRACE measuring changes in the Earth's gravitational field to do so



Rivers and lakes

- Rivers 2000 km³ and lakes 178000 km³ contain only 0.3% of the total freshwater but are the lifeline of human civilization and return water to the ocean. They also play a role in the rock cycle via sediment

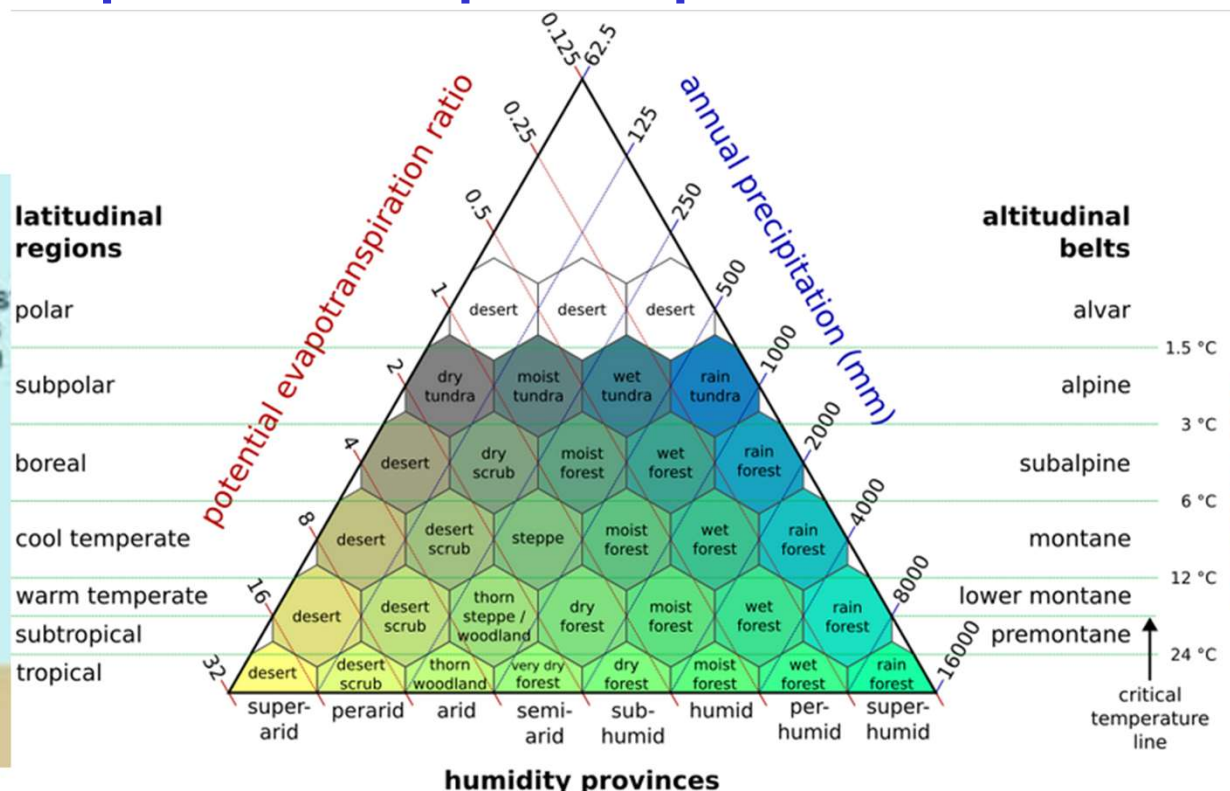
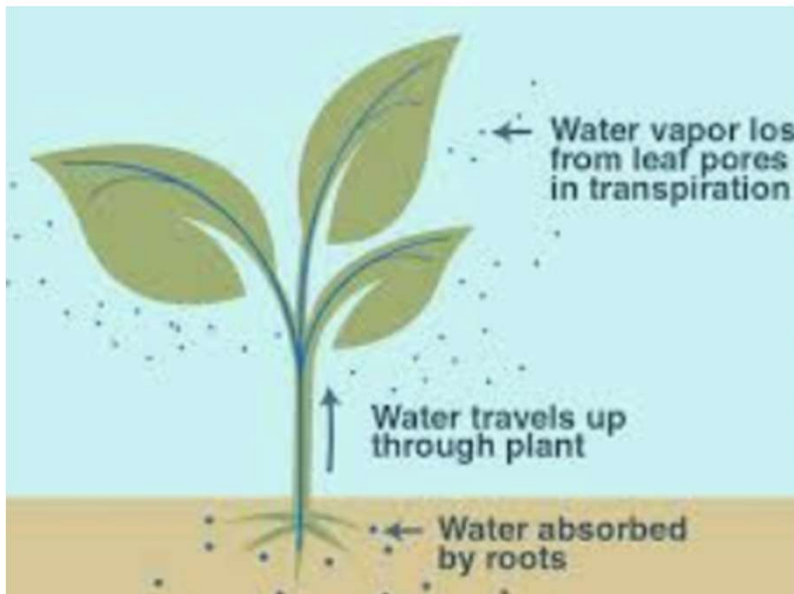


PRECIPITATION	The change of atmospheric water vapor to liquid (rain) or solid (snow)
EVAPORATION	The exchange of water from a liquid to a vapor
TRANSPIRATION	The release of water into the atmosphere by plant and animal cells
INFILTRATION	The movement of liquid water downward from the land surface into and through the soil and rock
RUNOFF	The total amount of water flowing in a stream

- Nutrient cycling in aquatic systems plays a key role in the phosphorous and nitrogen cycle

Soil moisture and biomass

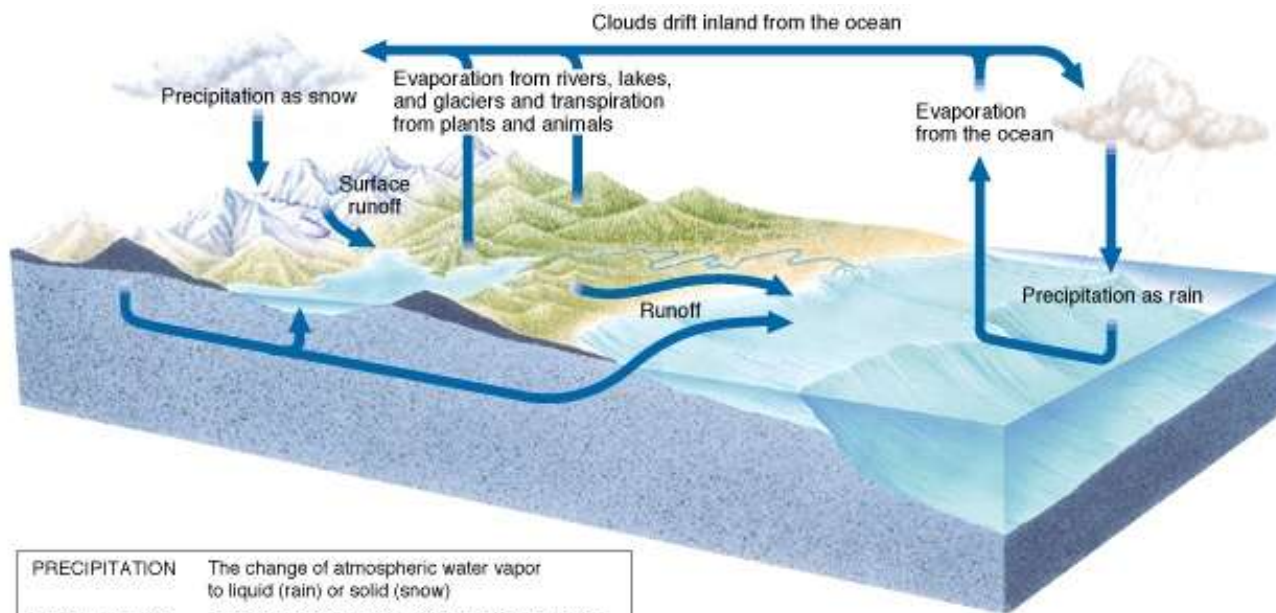
- The soil moisture 16500 km³ and biomass 1120 km³ reservoirs are closely interconnected.
- Plants continuously transpire water when stomata are open and the required water needs to be available in the form of soil moisture. How much water plants need to survive a given climate is calculated with the help of the temperature at the place and the rainfall seasonality of the rainfall. This quantity is called the potential evapotranspiration. The balance between rain and potential evapotranspiration determines the biome of a climate zone



Water Storage in the Atmosphere

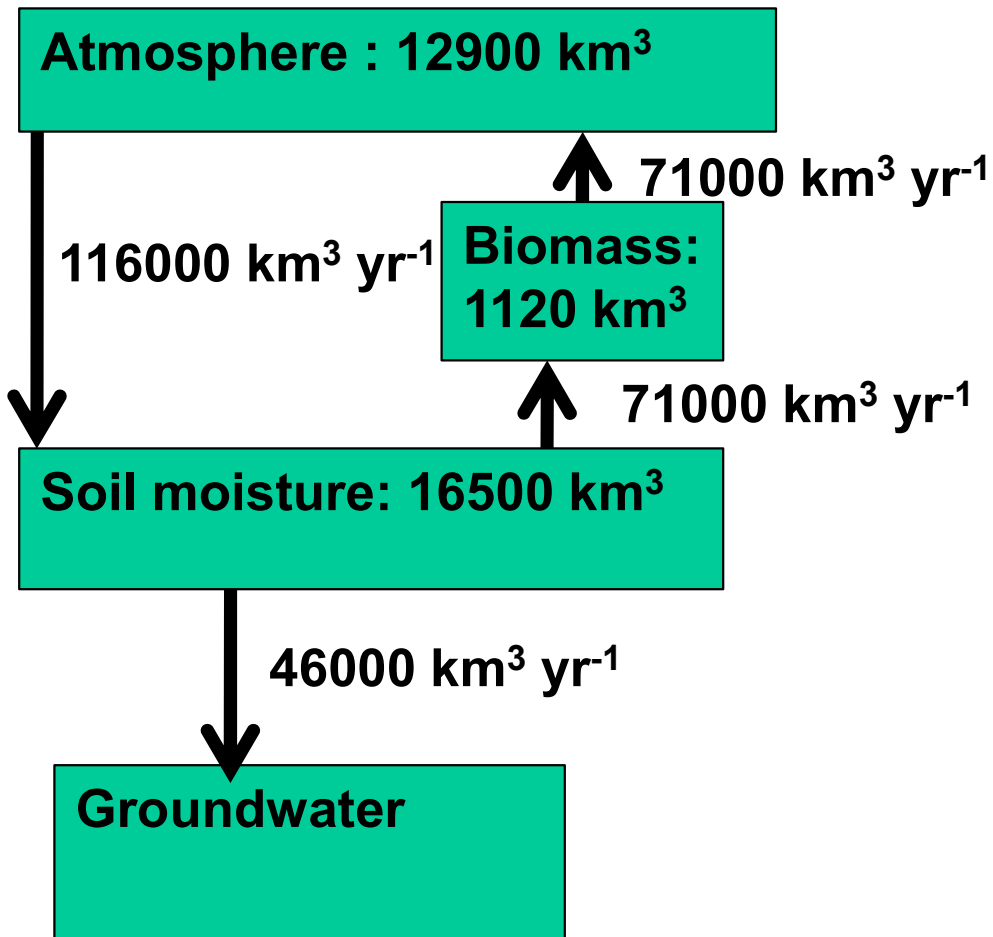
- Despite being the smallest reservoir containing 0.001% of the water only, the atmosphere plays a crucial role in moving moisture from the ocean to the land
- This reservoir is directly connected to all other reservoirs

- Clouds (water vapor condensed on particulate) play a key role in the albedo and in many biogeochemical cycles.



PRECIPITATION	The change of atmospheric water vapor to liquid (rain) or solid (snow)
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Numerical



Calculate the turnover time of the soil moisture reservoir in days.
Is the reservoir in steady state or not?